NATIONAL UNIVERSITY
of Computer & Emerging Sciences, Lahore

Department of Computer Science

CS4063 – Natural Language Processing
Fall 2026

Course outline prepared by: Dr. Maryam Bashir

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Course Information
Program: BSDS **Credit Hours:** 3 **Type:** Elective **Pre-requisites:** Linear algebra, Probability and Statistics, Programming in any high level language, Algorithms

Course Description:

Natural Language Processing (NLP) addresses fundamental questions at the intersection of human languages and computer science. Understanding complex language utterances is also a crucial part of artificial intelligence. Applications of NLP are everywhere because people communicate almost everything in language: web search, advertisement, emails, customer service, language translation, radiology reports, etc. There are a large variety of underlying tasks and machine learning models behind NLP applications. This course addresses topics like how computers can do useful things with human languages, e.g. filter junk email, extract social networks from the web, and find the main topics in the day's news. This course is also about how computational methods can help linguists explain language phenomena, including automatic discovery of different word senses and phrase structure.

Program Learning Outcomes (PLOs)

This course covers the following PLOs:

PLO #	PLO Name	PLO Description

2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the 16 abstraction and conceptualization of computing models from defined problems and requirements.
4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

Course Learning Outcomes (CLOs):

CLO #	CLO description	BT Domain/ BT Level	PLO #
1	Identify techniques for language modeling, smoothing methods, text classification	**C2 (Understanding)**	**PLO 2**
2	Represent documents and words as vectors for a variety of NLP tasks	**C2 (Understanding)**	**PLO 2**
3	Apply deep learning techniques for sequence modeling, machine translation and natural language generation	**C3 (Applying)**	**PLO 4**
4	Learn about transformer-based models and attention mechanism	**C3 (Applying)**	**PLO 4**
5	Learn about prompt engineering and large language models	**C3 (Applying)**	**PLO 4**

Course Textbook

Jurafsky and Martin, *SPEECH and LANGUAGE PROCESSING: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, Third Edition*, McGraw Hill. [Available Online]

Additional references and books related to the course:

1. Chris Manning and Hinrich Schütze, *Foundations of Statistical Natural Language Processing*, MIT Press. Cambridge, MA: May 1999.Steven Bird, Ewan Klein, and Edward Loper.
2. *Natural Language Processing with Python– Analyzing Text with the Natural Language Toolkit*. The first edition of the book, published by O'Reilly, is available at http://nltk.org/book_1ed/
3. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. **Attention is all you need**. In Proceedings of *the 31st International Conference on Neural Information Processing Systems* (NIPS'17)
4. Jacob Devlin,Ming-Wei Chang,Kenton Lee,Kristina Toutanova, **BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding**, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*
5. A Primer on Neural Network Models for Natural Language Processing By Yoav Goldberg https://arxiv.org/abs/1510.00726

Weekly Schedule

Week	Topics Covered	Text Book Chapter
1	Introduction, Motivation of Course, Text Preprocessing, Statistical Language Modeling, N Grams,	2, 3
2	Smoothing, Perplexity, Vector Semantics, TF-IDF, Cosine Similarity (Sparse Vectors), representations,	2,3, 6
3	Word Embeddings (WordtoVec) (Dense Vectors)	6, 4
4	Neural Language Modeling, Recurrent Neural Networks, Sequence Processing with Recurrent Networks	7, 9
5	Vanishing Gradient Problem	9
Midterm 1		
6	LSTM, Encoder Decoder Model, Neural Machine Translation, Natural Language Generation (NLG), Decoding schemes	13

7	Attention in encoder decoder model, NLG Evaluation Measures (ROUGUE, BLEU), Summarization	13
8	Self-Attention and Transformers,	10
9	Fine tuning and Masked Language Models	11
10	Pretraining, Prompting, Reinforcement Learning from Human feedback	11
	Midterm 2	
11	Sequence Labeling for Parts of Speech Tagging,	
12	Named Entities, Parsing	8
13	Presentations	
14	Presentations	

Tentative Grading Criteria

1. 3 to 4 Assignments (10%)
2. 3 to 4 Quizzes (10%)
3. Project (5%)
5. Midterm Exams (30%)
6. Final Exam (45%)

Course Policies

1. Quizzes will be announced.
2. No makeup for missed quiz or assignment.
3. 80% attendance

Plagiarism in Assignments

In writing up your assignments and in answering questions in exams, be as clear, precise, and as concise as you can. **Understandability will be an important factor in grading**.

Academic Integrity: All work MUST be done individually. Any copying of work from other person(s) or source(s) (e.g. the Internet) will automatically result in at least an F grade in the course. It does not matter whether the copying is done in an assignment, quiz, midterm exam, or final exam, it will be considered equally significant.